



# Optical timing jitter due to atmospheric turbulence: comparison of frequency comb measurements to predictions from micrometeorological sensors

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**Abstract:** During propagation through atmospheric turbulence, variations in the refractive index of air cause fluctuations in the time-of-flight of laser light. These timing jitter fluctuations are a major noise source for precision laser ranging, optical time transfer, and long-baseline interferometry. While there exist models that estimate the turbulence-induced timing jitter power spectra using parameters obtainable from conventional micrometeorological instruments, a direct and independent comparison of these models to measured timing jitter data has not been done. Here we perform this comparison, measuring turbulence-induced optical pulse timing jitter over a horizontal, near-ground path using frequency comb lasers while independently characterizing the turbulence along the path using a suite of micrometeorological sensors. We compare the power spectra of measured optical pulse timing jitter to predictions based on the measured micrometeorological data and standard turbulence theory. To further quantitatively compare the frequency comb data to the micrometeorological measurements, we extract and compare the refractive index structure parameter,  $C_n^2$ , from both systems and find agreement to within a factor of 5 for wind speed  $>1$  m/s, and further improvement is possible as wind speed increases. These results validate the use of conventional micrometeorological instruments in predicting optical timing jitter statistics over co-located laser beam paths.

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## 1. Introduction

Optical turbulence in the atmosphere arises due to the turbulent nature of the air which results in eddies of variable refractive index [1]. As light propagates through the atmosphere, optical turbulence modulates its intensity and phase. Over the past 60 years, the theoretical and experimental attention of the optical turbulence community has focused on understanding intensity fluctuations, often referred to as scintillation, due to their particularly deleterious effects on optical imaging and free-space optical communication [2,3]. In efforts to improve astronomical observation, the field of adaptive optics additionally grew our understanding of atmospheric phase fluctuations in the higher order aberrative modes (tilt, defocus, astigmatism, etc.) [4–6]. In part because it is not important for adaptive optics, the lowest order phase fluctuation, called the piston mode, has experienced comparatively less experimental attention.

The piston mode aberration manifests as phase noise or timing jitter, and is an important noise source in laser ranging [7–9], long-baseline and stellar interferometry [10,11], and free-space optical time transfer [12–18]. While theoretical descriptions of atmospheric phase noise have

existed since the publication of Tatarskii's work in 1961 [19], experimental characterizations of phase fluctuations over wide length scales have proven to be challenging.

Early measurements of turbulent fluctuations of optical phase in the 1970s investigated the differential phase power spectrum and phase structure function by interfering two points from the same continuous-wave laser beam [20–23]. To directly measure phase fluctuations rather than differential phase, the traditional approach has been to use continuous wave interferometry [8,10,24–26]. While these interferometric experiments are highly sensitive, the fringes must be continuously visible to track phase changes over time. Loss of beam power due to scintillation, beam steering, or beam obstruction could therefore cause phase slips, reducing the duration of phase-continuous measurements.

The invention of the optical frequency comb has enabled hours-long measurements that do not suffer from  $2\pi$  phase ambiguities after full signal loss [27]. Using frequency combs and linear optical sampling to measure the timing jitter of pulses, researchers have measured turbulence-induced phase noise across a 2 km outdoor slant-path, finding order-of-magnitude agreement to theoretical models [13]. However, the phase power spectra showed reduced power laws in the inertial subrange and contained no evidence of the turbulence outer scale. These measurements were conducted over complex terrain including multiple buildings and a steeply sloped hill. This topography coupled with the structures could very well have induced non-homogenous turbulence along the beam path. Furthermore, no micrometeorological sensors were deployed along the propagation path, so the atmospheric parameters in the models could not be independently determined. Using mode-locked lasers and balanced optical cross correlation, other researchers have measured timing jitter due to indoor atmospheric turbulence and found reasonable agreement to the Kolmogorov power law between 100 mHz and 10 Hz [28].

To build on the previous work described above, we measured optical timing jitter due to atmospheric turbulence across a uniform outdoor path while independently characterizing the turbulent conditions using co-located micrometeorological sensors in the form of ultrasonic anemometer-thermometers (hereafter referred to as “sonics”) and a quartz-crystal barometer. We use the data from the micrometeorological sensors to empirically determine the free parameters in three theoretical models for the timing jitter power spectral density and compare these models to the optical timing spectra, finding the best agreement to the Greenwood-Tarazano model [1,29]. We also quantitatively compare the refractive index structure function,  $C_n^2$ , derived from each the optical timing and micrometeorological data, finding agreement at moderate to high wind speeds ( $>1$  m/s). These results show that micrometeorological data can help estimate timing jitter across nearby optical paths with the caveat that lower wind speeds will cause systematic under-prediction of the timing jitter power spectrum at all frequencies.

The structure of this paper is as follows. In Sec. 2 we present the established theory of turbulence-induced optical timing jitter. Next in Sec. 3, we present the experimental setup and in Sec. 4 we present the results from a weeklong measurement campaign. In Sec. 5 we compare these results to theory and present a direct comparison of  $C_n^2$  extracted from the frequency comb and micrometeorological sensor measurements. Finally, in Sec. 6 we conclude with a summary of our findings and consider their implications.

## 2. Theory

In this section we outline several models for atmospheric timing jitter following the work of Tatarskii [30]. These models will be used with  $C_n^2$  and wind speed input from the micrometeorological instruments for a direct comparison with the timing jitter measured with the frequency combs. In Sec. 2.1 we begin with a discussion of the 3D spatial spectrum of optical turbulence. Then, in Sec. 2.2 we outline the derivation used to calculate the path-integrated temporal turbulence spectrum from the spatial spectrum. Finally, in Sec. 2.3, we present the

models discussed in Sec. 2.1 reformulated as functions of time, not space, along an integrated path. Ultimately, it will be these temporal spectra that we compare directly to the data.

### 2.1. Turbulence spectra

The refractive index of air is a function of the air temperature, pressure, and moisture content [31]. Turbulent atmospheric mixing creates eddies of varying refractive indices. For isotropic turbulence the statistics of these eddies are described by the spatial power spectrum of the refractive index,  $\Phi_n(\kappa)$ , where the independent variable  $\kappa$  is the magnitude of the three-dimensional wavenumber  $\kappa = (\kappa_1^2 + \kappa_2^2 + \kappa_3^2)^{1/2}$  [rad/m]. The simplest model of this spectrum is the so-called Kolmogorov model given by:

$$\Phi_n(\kappa) = 0.033 C_n^2 \kappa^{-11/3}, \quad 1/L_0 < \kappa < 1/l_0, \quad (1)$$

where the bounds are set by the outer and inner scale lengths,  $L_0$  and  $l_0$  respectively. Physically these lengths can be thought of as the largest and smallest turbulent eddy size where energy is injected and dissipated in a turbulent system [1].

To loosen the length scale restrictions on  $\kappa$ , several alternatives to Eq. (1) have been developed that account for the effects of the outer scale with filter functions. One such spectrum is the von Kármán spectrum [32]:

$$\Phi_n(\kappa) = 0.033 C_n^2 [\kappa^2 + (1/L_0)^2]^{-11/6}, \quad 0 < \kappa < 1/l_0. \quad (2)$$

This spectrum rolls off from the  $-11/3$  power law to a constant magnitude at frequencies below  $\kappa < 1/L_0$ . Note we have omitted the exponential term in Eq. (2) that accounts for the inner scale effects to simplify this analysis. The frequency comb system noise floor obscures the turbulence spectrum at the higher Fourier frequencies making measurements of the inner scale inaccessible.

While Eq. (2) presents a convenient analytical form, experimental work over several decades has shown this spectrum often underestimates the energy at low wavenumbers [7,11,29]. This prompted Greenwood *et al.* [29] to propose a new, empirically-derived spectrum based on microtemperature measurements near the ground, called the Greenwood-Tarazano spectrum:

$$\Phi_n(\kappa) = 0.033 C_n^2 [\kappa^2 + (\kappa/L_0)]^{-11/6}, \quad 0 < \kappa < 1/l_0, \quad (3)$$

The Greenwood-Tarazano model exhibits a more gradual rolloff below  $\kappa = 1/L_0$  than the von Kármán spectrum. Physically this indicates that length scales larger than the nominal outer scale still influence the timing jitter [1].

It should be noted that there are different conventions in the literature regarding the exact forms of Eqs. (2) and (3), with some sources [1,22,31] using  $2\pi/L_0$  in place of  $1/L_0$ . Here we have chosen the latter convention used in Refs. [13,29,32] to maintain consistency with the Greenwood-Tarazano spectrum originally proposed in [29].

### 2.2. Conversion from spatial spectra to path integrated temporal spectra

Whereas Eqs. (1), (2), and (3) define the spatial statistics of volumetric turbulence, the timing jitter data generated by our frequency comb system is in time series form and measures timing fluctuations accumulated as the light propagates. We therefore need a temporal power spectrum that accounts for integrated fluctuations through three-dimensional turbulence for a direct comparison.

Tatarskii derives the temporal spectrum of phase fluctuations accumulated during propagation [30, Ch. 4, Sec. 52]. Specifically, Eq. (52.33) from Ref. [30] gives the temporal spectrum as the

integral of the 2D spectrum:

$$S_\phi(f) = 8\pi/V_\perp \int_0^\infty F_\phi(\kappa_\perp, L, k) d\kappa_1, \quad (4)$$

where  $F_\phi$  is the 2D spectrum which is related to the 3D spectrum by a Fourier transform in one dimension [30, pg. 30],  $\kappa_\perp = (\kappa_1^2 + \kappa_2^2)^{1/2}$  is the 2D wavenumber perpendicular to the beam path,  $L$  is the length of the propagation path,  $k$  is the optical wavenumber, and  $V_\perp$  is the magnitude of the wind vector perpendicular to the beam path.  $F_\phi$  is given in Ch. 3, Sec. 46, Eq. (37a) in Ref. [30] as

$$F_\phi(\kappa_\perp) = \pi k^2 L g(\kappa_\perp) \Phi_n(\kappa_\perp), \quad (5)$$

where  $g(\kappa_\perp, L)$  is a filter function to account for diffraction written as

$$g(\kappa_\perp) = 1 + \frac{k}{\kappa_\perp^2 L} \sin\left(\frac{\kappa_\perp^2 L}{k}\right). \quad (6)$$

At the length scales accessible to our measurements,  $(\kappa_\perp^2 L)/k \ll 1$ , so we can approximate  $g(\kappa_\perp, L) \approx 2$ . It is also worth noting that replacing the scalar 3D wavenumber,  $\kappa$ , with the 2D wavenumber,  $\kappa_\perp$ , in  $\Phi_n$ , as was done for Eq. (5), is only valid if we assume locally isotropic turbulence.

To convert to a temporal spectrum, we must also apply Taylor's Frozen Turbulence Hypothesis, which posits that turbulent eddies remain unchanged within the time it takes for wind to move them laterally across the beam path [33]. This allows us to equate a single spatial frequency orthogonal to the beam path, to a temporal frequency according to  $\kappa_2 = 2\pi f/V_\perp$  where  $f$  is a temporal frequency in Hz. Using this assumption as well as the locally isotropic assumption described above we can substitute Eq. (5) into Eq. (4) to get the one-sided spectrum:

$$S_\phi(f) = \frac{16\pi^2 k^2 L}{V_\perp} \int_0^\infty \Phi\left(\sqrt{\kappa_1^2 + (2\pi f/V_\perp)^2}\right) d\kappa_1. \quad (7)$$

### 2.3. Phase and timing jitter power spectral density

Having shown the steps for switching between spatial and temporal spectra, we now present temporal phase and timing jitter models associated with the Kolmogorov, von Kármán and Greenwood-Tarazano spectra. Substituting Eq. (1) for  $\Phi_n$  in Eq. (7) we get the Kolmogorov model for the phase power spectral density, equivalent to Ch. 4, Sec. 52, Eq. (35) in [30]:

$$S_\phi(f) = 0.0326 k^2 C_n^2 L f^{-8/3} V_\perp^{5/3}, \quad (8)$$

Similarly, the von Kármán spectrum, Eq. (2), gives

$$S_\phi(f) = 4.384 \frac{k^2 C_n^2 L}{V_\perp} \left( \frac{4\pi^2 f^2}{V_\perp^2} + \frac{1}{L_0^2} \right)^{-4/3}, \quad (9)$$

and the Greenwood-Tarazano spectrum, Eq. (3), gives

$$S_\phi(f) = 5.211 \frac{k^2 C_n^2 L}{V_\perp} \int_0^\infty \left[ \frac{4\pi^2 f^2}{V_\perp^2} + \kappa_1^2 + L_0^{-1} \left( \frac{4\pi^2 f^2}{V_\perp^2} + \kappa_1^2 \right)^{1/2} \right]^{-11/6} d\kappa_1. \quad (10)$$

To directly compare to our data, we must convert Eqs. (8), (9), and (10) from phase spectra to timing jitter spectra using a dimensional term:

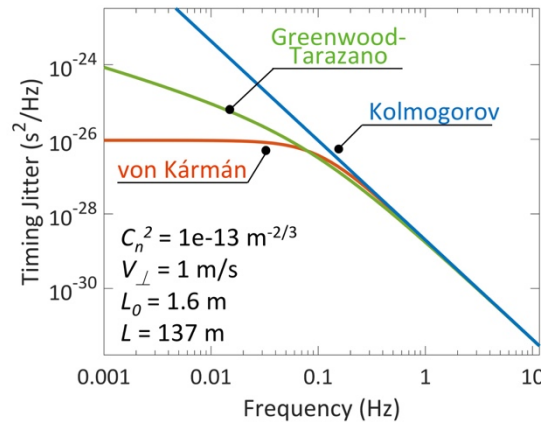
$$S_{jitter} = \left( \frac{n_g \lambda_0}{2\pi c} \right)^2 S_\phi(f), \quad (11)$$

where  $n_g$  is the group refractive index approximated to 1 hereafter,  $\lambda_0 = 2\pi/k$  is the central wavelength, and  $c$  is the speed of light.

Finally, to account for correlated turbulence along our folded (out-and-back) experimental path (discussed in Sec. 3), we multiply the temporal spectrum by a factor of 4. The factor of 4 comes from treating the timing jitter accumulated over the path to and from the retroreflector as two separate random variables,  $X$  and  $Y$ , and the total measured timing jitter for the round trip as the sum of the two,  $Z = X + Y$ . Since  $\text{var}(Z) = \text{var}(X) + \text{var}(Y) + 2\text{cov}(X, Y)$  and the turbulence is perfectly correlated over the timescales of the pulse propagation,  $\text{cov}(X, Y) = \text{var}(X)$  and  $\text{var}(Z) = 4\text{var}(X)$ . Variance is the integral of the power spectrum so we can multiply the one-way temporal spectrum by 4 to get the two-way temporal spectrum. For the Kolmogorov spectrum this gives the final form of  $S_{\text{jitter}}$ :

$$S_{\text{jitter}}(f) = (4)(0.0326)c^{-2}L C_n^2 f^{-8/3} V_{\perp}^{5/3}. \quad (12)$$

Equation (12) is shown in Fig. 1 alongside the equivalent  $S_{\text{jitter}}$  forms of the von Kármán and Greenwood-Tarazano models. Hereafter we refer to these  $S_{\text{jitter}}$  models as the Kolmogorov, von Kármán, and Greenwood-Tarazano models respectively.



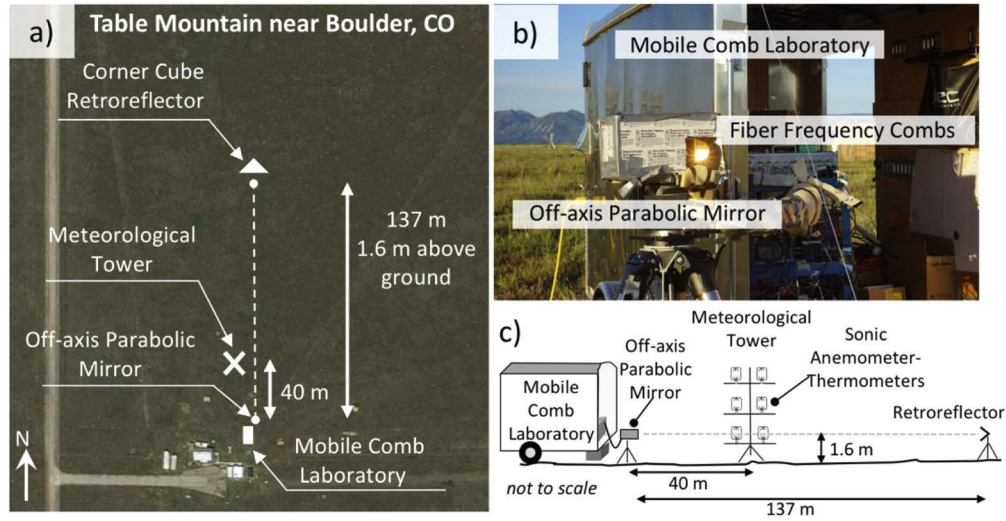
**Fig. 1.** Comparison of timing jitter power spectral density models,  $S_{\text{jitter}}$ . Notice the low-frequency variation between  $S_{\text{jitter}}$  calculated using the Kolmogorov (blue), Greenwood-Tarazano (green) and von Kármán (red) spectra.

### 3. Experimental setup

The experiment was set up in a field at the Department of Commerce Table Mountain Test Facility 10 miles north of Boulder, Colorado, shown in Fig. 2(a). The field was nearly flat and covered with a mixture of soil, stone, and grass interspersed with low shrubs. The mobile comb laboratory, pictured in Fig. 2(b), housed most of the equipment including the combs and their control electronics. Light launched from an off-axis parabolic mirror just outside the mobile comb lab was directed to a retroreflector 137 m away. To characterize the turbulence with conventional micrometeorological instruments, we set up a meteorological tower approximately 40 meters from the mobile laboratory, about 2 m from the beam path. While the trailer hosting the laboratory surely disrupted the inherent turbulence conditions of the otherwise minimally featured field, the need to minimize and stabilize the optical fiber connecting the combs and free-space launching optics limited the separation between them.

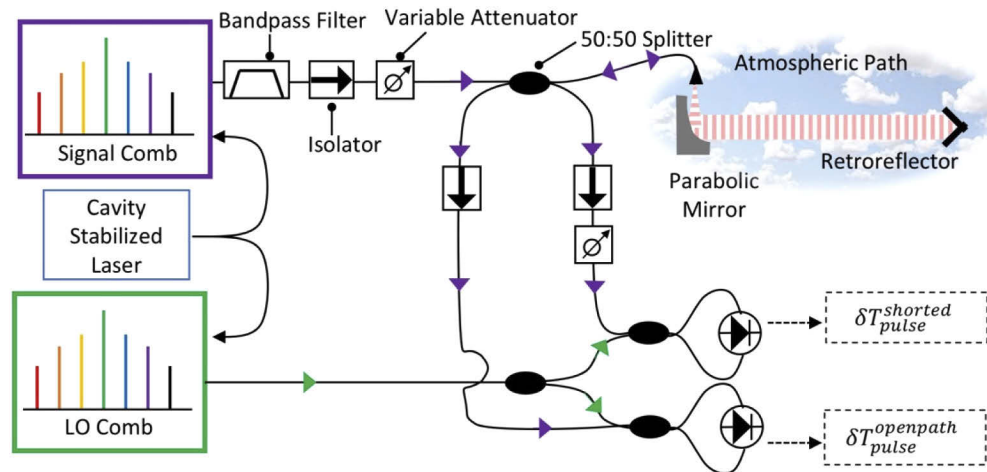
#### 3.1. Frequency comb laser measurements

To measure the atmospheric timing jitter, we used two self-referenced optical frequency comb lasers (“combs”), phase-locked to a field-deployable cavity stabilized laser. The combs used here



**Fig. 2.** Experimental setup. a) Aerial diagram of the experiment at Table Mountain showing the orientation of the laser path to the mobile comb laboratory, meteorological tower, and surrounding buildings. b) Photograph showing the mobile comb laboratory housing the fiber frequency combs and the relative location of the off-axis parabolic mirror collimator. c) Side-view diagram of the experiment showing the elements in a).

follow Ref. [34], with pulse repetition rates of approximately 200 MHz. One laser is designated as the signal comb and the other as the local oscillator or “LO” comb. We split the light from the signal comb into two branches, both ultimately bound for heterodyne detection with the LO comb. We sent half the light over a shorted fiber branch and the other half over the atmospheric link that consisted of an off-axis parabolic mirror collimator to launch and receive the light and a retroreflector. We band-pass filtered the light from the signal comb to 14nm centered at 1560nm for both branches. A schematic of this experimental setup is shown in Fig. 3.



**Fig. 3.** Schematic of experimental setup to measure atmospheric timing jitter  $\delta T_{atmosphere}$ . Both the signal and local oscillator frequency combs are phase-locked to the cavity-stabilized laser. Colored arrows indicate the direction light over single-mode PM fiber (black lines). The free-space portion of the setup is shown in striped red and white.

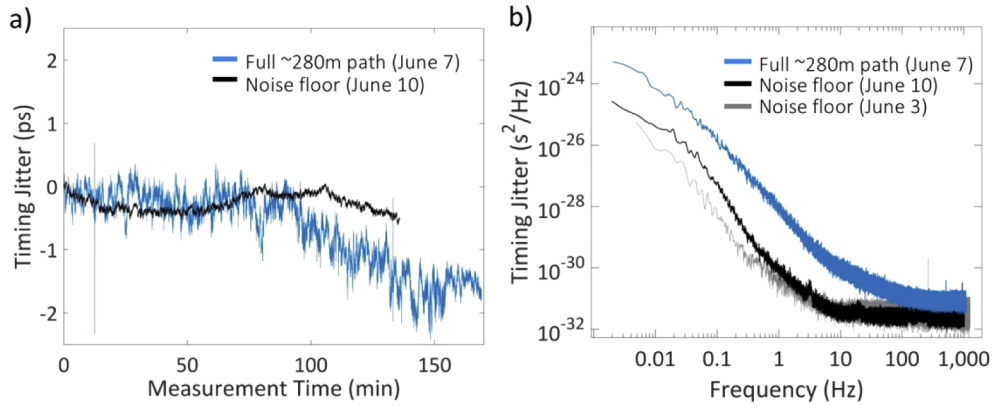
The pulses of the signal comb were detected using linear optical sampling [35,36]. By detuning the repetition rate local oscillator comb ( $f_{r,LO}$ ) to be  $\sim 2$  kHz higher than the repetition rate of the signal comb ( $f_{r,S}$ ), the mixed light from the two combs formed interferogram signals repeating every  $1/(f_{r,LO} - f_{r,S}) = 1/(\Delta f_r)$  seconds as the pulses from the two combs stepped through one another. The generated interferogram signals can be interpreted as down-sampled versions of the signal comb pulses. The points comprising the interferograms are measured by balanced detection and the interferogram arrival time (nominally given at the occurrence of the interferogram's peak amplitude) is extracted as described in Ref. [37]. Linear optical sampling allows us to map timing jitter in these interferogram arrival times,  $\delta t_{IGM}$ , to the timing jitter in the arrival of time of the signal pulses,  $\delta T_{pulse}$ , (with respect to the LO comb pulses) according to [13]

$$\delta T_{pulse} = (\Delta f_r / f_{r,S}) \delta t_{IGM}. \quad (13)$$

We simultaneously measure the timing jitter over a fiber-shortened path, denoted  $\delta T_{pulse}^{shorted}$ , and over atmospheric open path, denoted  $\delta T_{pulse}^{openpath}$ . Their difference is insensitive to timing jitter from the combs themselves and from most path length variations within the detection system, and yields the atmospheric timing jitter as

$$\delta T_{atmosphere} = \delta T_{pulse}^{openpath} - \delta T_{pulse}^{shorted}. \quad (14)$$

From Fig. 3, it is clear the fiber shortened branch is only partially common to the atmospheric-bound branch, so we measured the entire-system noise floor both before and after the measurement campaign by placing the retroreflector  $< 1$  m from the parabolic mirror and enclosing the air between the two to minimize turbulence. Figure 4 compares the timing jitter measured over the full 280-m optical path to this system noise floor. There is an overall system drift on the order of 0.5 picoseconds over 2.5 hours that is likely due to a combination of the tripods settling and temperature variations seen by the optical fiber despite efforts at insulation. Nevertheless, the noise floor PSD is more than a decade below the typical atmospheric piston noise for Fourier frequencies from 10 Hz to below 10 mHz.



**Fig. 4.** a) Time series of system noise floor timing jitter (black) shown with atmospheric timing jitter (blue) for scale. b) Corresponding power spectral densities for the data in a) of system noise floor measurements (grey and black) and a typical atmospheric measurement (blue).

We should also mention that a cycle slip in the phase lock of either comb would result in 5 fs jumps in the timing data. While we see no evidence of cycle slips in our data, we cannot verify their absence during moments of signal loss due to scintillation. During these times we rely on the low probability of cycle slips in our system, which has demonstrated continuous slip-free operation over several weeks [27,38].

### 3.2. Micrometeorological measurements

Four ultrasonic anemometers-thermometers (“sonics”) and one quartz-crystal barometer were deployed approximately 40 m north of the frequency-comb system and 2 m west of the propagation path. The sonics were mounted on a 6 m tall portable meteorological tower. The sonic used to measure the wind vector and determine  $C_n^2$  for this work had the center of its measurement volume located at  $z_1 = 1.59$  m, approximately the height of the laser beam path. The setup of the sonics was similar to earlier optical-propagation field studies [39–41]. The barometer was connected to a Quad Disk [42], a device that mitigates wind noise in the measurements of the static pressure.

The sonics provided time series of air temperature and wind velocity at a rate of 32 Hz. According to the manufacturer, the wind measurements have a precision of 1 cm/s and an absolute accuracy of 5 cm/s. The precision of the temperature measurements is 0.01 K, and the absolute accuracy is better than 2 K.

In this paper we consider three wind velocity components:  $V_{\parallel}$  the component parallel to the laser beam path,  $V_{\perp,H}$ , the horizontal component perpendicular to the laser beam path, and  $V_{\perp,V}$ , the vertical component which is always perpendicular to the laser beam path. The magnitude of the perpendicular wind vector,  $V_{\perp}$  introduced in Sec. 2.2, can be written using these components according to  $V_{\perp} = (V_{\perp,H}^2 + V_{\perp,V}^2)^{1/2}$ .

The barometer was set up to sample at a sampling rate of 10 Hz. According to the manufacturer, its absolute accuracy is 10 Pa (0.01% of the atmospheric pressure). In a study on atmospheric infrasound [43], the precision of the 10 Hz samples were verified to about 1  $\mu$ bar (0.1 Pa).

We used the micrometeorological measurements to retrieve the refractive index structure parameter,  $C_n^2$ , through the relationship

$$C_n^2 = \frac{\langle [n(t + \tau) - n(t)]^2 \rangle}{(U\tau)^{2/3}}, \quad (15)$$

where

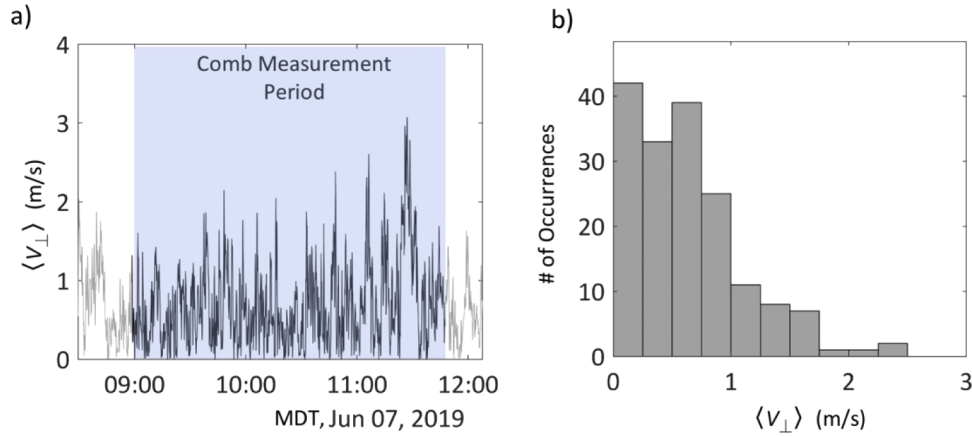
$$n(t) = 1 + a \frac{\langle P(t) \rangle}{T(t)} \quad (16)$$

is the refractive index of air,  $P$  is the static air pressure in units of Pa,  $T$  is absolute temperature in K,  $U = (\langle V_{\perp,H}^2 \rangle + \langle V_{\parallel}^2 \rangle)^{1/2}$  is the time-averaged magnitude of the mean horizontal wind vector,  $\tau$  is a time lag,  $a = 7.8 \times 10^{-7}$  K/Pa is a constant [44, p. 10], and the angle brackets represent the time average over 1 min. We chose  $\tau$  such that for each 1 min interval, the spatial separation (by invoking Taylor’s frozen-turbulence hypothesis) was  $r = U\tau = 1$  m. This value of  $r$  was chosen to be small compared to  $2z_1 = 3.18$  m, the diameter of the largest isotropic turbulent eddies centered at  $z_1$  [45], and large compared to the size of the sonic’s sampling volume of about 10 cm. For  $r = 1$  m, a typical wind speed of  $U = 1$  m/s corresponds to a time lag of  $\tau = 1$  s.

## 4. Results

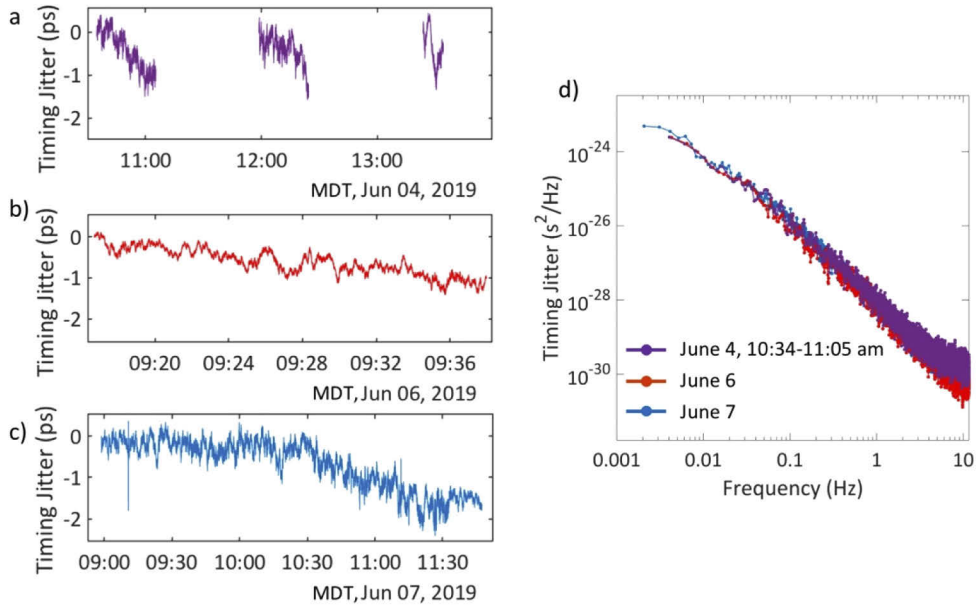
During the weeklong measurement campaign, we gathered around 4.5 hours of frequency comb timing jitter data, with the longest continuous period being 169 minutes. The optical timing jitter data was taken between 8 am and 12:30 pm MDT on three consecutive days as thunderstorms prevented afternoon measurements. The air temperature, averaged over 20 min, ranged from 8 °C before sunrise on June 1 to 27 °C in the afternoon of June 7. Winds were light and variable, with typical wind speeds ranging between 0 and 2 m/s as shown in Fig. 5.

Figure 6 shows the time series of the turbulence induced timing jitter,  $\delta T_{\text{atmosphere}}$ . In the longest measurement, taken on June 7<sup>th</sup>, we detect  $\sim 2$  ps of variation in the time of flight over almost three hours with higher frequency structure in the signal as expected from turbulence.



**Fig. 5.** Wind speeds during the comb measurement period on June 7. a) Time series of 1 minute averages of the absolute value of the perpendicular wind speed vector,  $\langle V_{\perp} \rangle$ . (b) Histogram of  $\langle V_{\perp} \rangle$  for the time series shown in a).

The received optical power varied from below 2 nW (our detection threshold) up to  $\sim 60$   $\mu$ W because of scintillation.

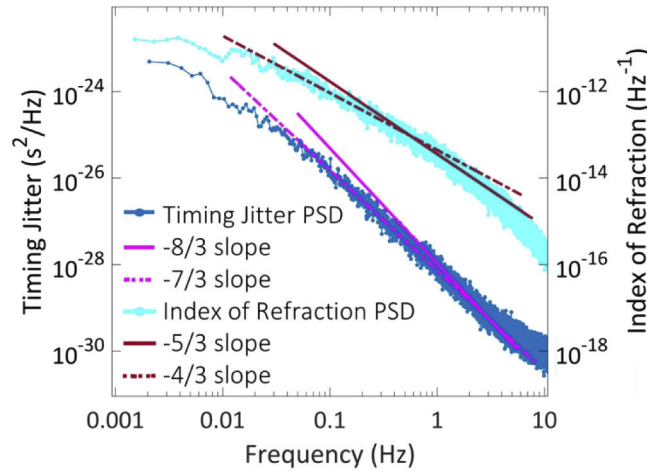


**Fig. 6.** Time series of  $\delta T_{atmosphere}$  measured over the 280 m path. The starting value of each phase-continuous measurement is set to 0 ps. a) Measurements on June 4<sup>th</sup> 2019. b) Measurements on June 6<sup>th</sup> 2019. c) Measurements on June 7<sup>th</sup> 2019.

#### 4.1. Characteristics of timing jitter power spectral density

To compare our results to the theory described in Sec. 2, we estimate power spectral densities (PSDs) from the  $\delta T_{atmosphere}$  time series using Welch's method of averaged, modified periodograms

[46] with a Hamming window. A representative timing jitter PSD estimated from Fig. 6(c) is shown in Fig. 7.



**Fig. 7.** Power spectral densities of the timing jitter, taken with the frequency comb system, (dark blue) and index of refraction variation, derived from the sonic anemometer-thermometer and barometer data (teal). The time series used to create the spectra were taken concurrently. Also shown are -8/3 (solid pink) and -5/3 (solid red) power laws predicted by theory in Ref. [30] for the timing jitter and index respectively. Since we see an apparent reduction in the power-law of the timing jitter data we show the -7/3 power law (dashed pink) on the timing jitter data along with the similarly reduced -4/3 power law (dashed red) on the index data. The curves are vertically offset for clarity and power law lines are a visual aid, not lines of best fit.

For frequencies,  $f$ , below 5 Hz, the PSD approximately follows the -8/3 power law predicted by theory before rolling off at  $\sim 0.1$  Hz. However, careful examination of the slope in the  $0.1 \text{ Hz} < f < 5 \text{ Hz}$  region shows an apparent reduction in the power-law exponent. For the spectrum shown in Fig. 7, the line of best fit is  $\propto f^{-6.7/3}$ . This is consistent with Ref. [13] and the review of similarly reduced slopes therein. The potential causes of this reduced slope are numerous. Reference [13] points to a breakdown of Taylor's frozen-turbulence hypothesis which might be possible for our measurements due to the low perpendicular wind speeds. For example, during the June 7<sup>th</sup> run we measured the standard deviation of  $V_{\perp}$  divided by its time average to be 1.09, which is much greater than the value 0.33 empirically set as the upper limit for the validity of Taylor's Hypothesis by Ref. [47].

We investigate the power law behavior of the index of refraction PSD where  $n(t)$  was calculated from Eq. (16), to see if it shares the same reduced slope as the frequency comb measurements. The same scaling laws that predicts a -8/3 power-law for the pulse timing would predict a -5/3 power-law for the index of refraction PSD in the inertial subrange [30]. We can see from Fig. 7 that indeed both PSDs show a reduction in their apparent power law at lower frequencies. The index spectrum follows a -5/3 power law until  $\sim 0.5$  Hz, where it rolls off to almost -4/3. Note that the finite length of the sonics' ultrasonic paths leads to a spatial low-pass filtering and therefore to a frequency spectrum steeper than -5/3 at greater than 3 Hz. As described above, the timing jitter data sees a reduced slope over most of its range.

From Fig. 4(a) we know the system noise floor begins to dominate our measurement above 100 Hz. When measuring over the full  $\sim 280$  m path, there is additional noise in the 10 Hz to 100 Hz region that is correlated with times when the return laser beam signal is temporarily lost. The timing jitter measurement is robust against temporary loss of signal, but the loss of signal still

introduces gaps in the timing jitter data ranging from  $\sim 500 \mu\text{s}$  to  $\sim 3 \text{ s}$  in length. To generate our power spectral densities, we need continuous time series and these gaps must be removed by either concatenation or interpolation of the data. Here, we interpolate and simulations show that this leads to an artificial increase in noise at frequencies above 10 Hz. We also believe some coupling between the angle-of-arrival and piston modes may be an additional source of noise above 5 Hz.

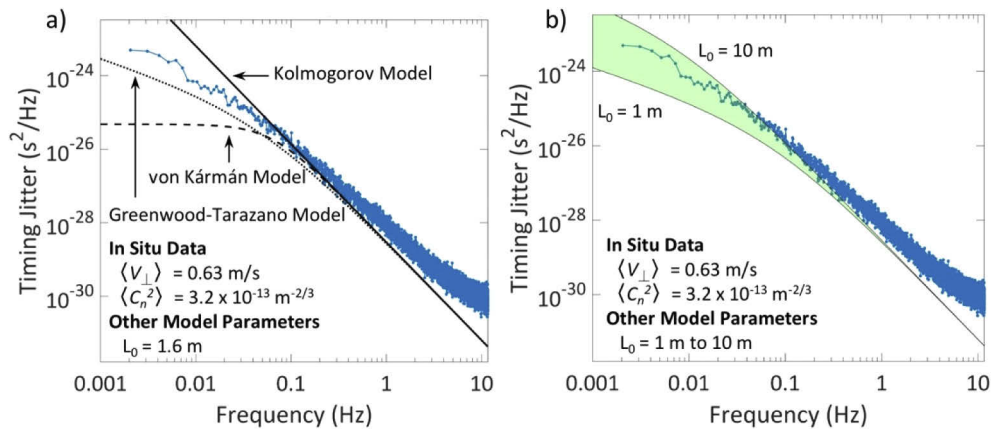
## 5. Discussion

As described in Sec. 2, there are multiple spectral models of timing jitter with varying low-frequency behavior. In Sec. 5.1 we compare these models to our data, and in Sec. 5.2 we establish a quantitative comparison between the frequency comb and micrometeorological measurements through a comparison of  $C_n^2$  extracted from each system.

### 5.1. Comparison to existing spectral models

We are interested in the agreement of the measured timing jitter spectra to theoretical curves generated with empirical parameters from the sonics. For the wind parameter we used the average perpendicular wind speed during the measurement period,  $\langle V_\perp \rangle$ . For the  $C_n^2$  parameter we used a time-averaged  $C_n^2$  value extracted from the micrometeorological data using the direct structure-function approach with a separation of 1 m. The only other free parameter in the models is  $L_0$  which we have set to the beam height above ground (1.6 m) [4,21,22,26].

We first compare to the Kolmogorov model. As shown in Fig. 8(a), this model predicts the timing jitter PSD to within a factor of 10 at 10 Hz dropping to a factor of  $\sim 2\times$  between 0.1 Hz and 1 Hz. In Sec. 2 we discussed two theoretical models that account for outer scale effects: the Greenwood-Tarazano model and the von Kármán model. Looking at the low-frequency ( $f < 0.1 \text{ Hz}$ ) region our data in Fig. 8(a), we see the gradual rolloff of the Greenwood-Tarazano better fits the data than the more dramatic von Kármán model. Figure 8(b) shows the effect of varying the outer scale value between 1 m and 10 m for the Greenwood-Tarazano model. These results indicate that use of the von Kármán model could lead to underestimation of timing jitter and timing variance over a turbulent path as described theoretically in Refs. [15–17].

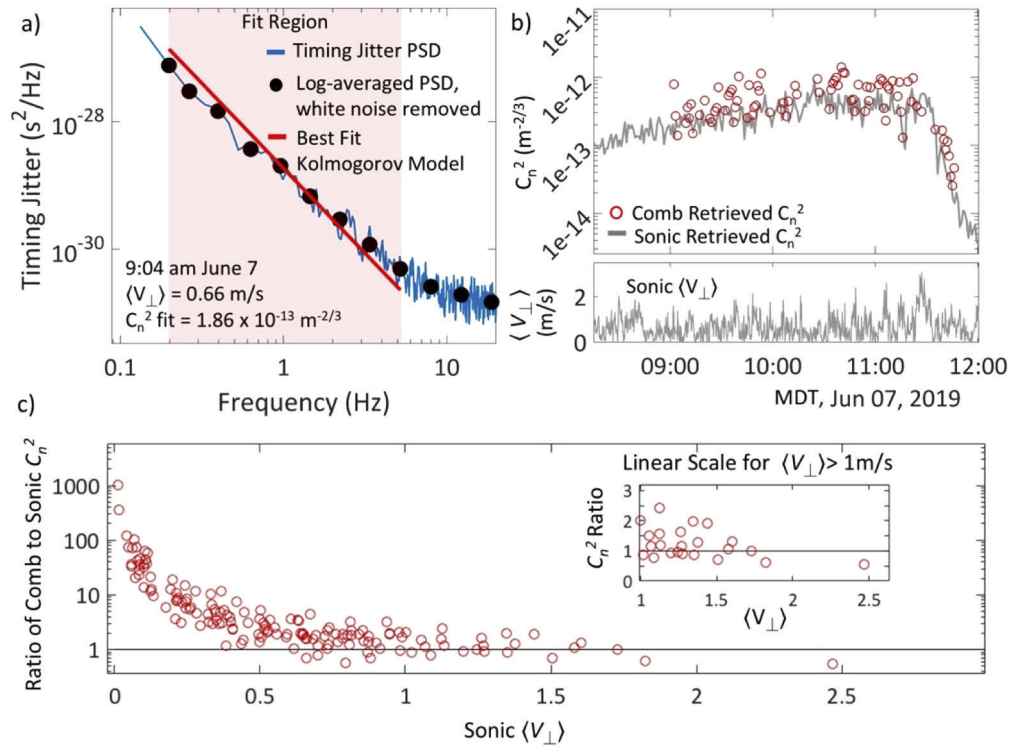


**Fig. 8.** a) Power spectral density of timing jitter from June 7<sup>th</sup> with Kolmogorov, von Kármán and Greenwood-Tarazano models using  $V_\perp$  and  $C_n^2$  from micrometeorological sensors and  $L_0$  chosen to be height above ground, 1.6 m. b) Same power spectral density showing only the Greenwood-Tarazano model for  $L_0$  ranging from 1 m to 10 m.

## 5.2. $C_n^2$ comparison

While we showed qualitative agreement between the models and optical timing jitter data in Sec. 5.1, to quantitatively compare the frequency comb and micrometeorological sensor turbulence measurements over time, it is simplest to consider a single parameter. For this comparison, we extract  $C_n^2$  from both datasets in one-minute time intervals.

To retrieve  $C_n^2$  from the optical timing jitter data, we generate PSDs for each minute of data. We then average the PSD over logarithmically spaced frequency bins. A typical 1-minute PSD (blue line) and corresponding log-spaced average points (black dots) are shown in Fig. 9(a). The PSD is fit to the Kolmogorov model over the frequency range  $0.19 \text{ Hz} < f < 5 \text{ Hz}$  with the wind speed fixed from the sonic data and only  $C_n^2$  varied. We chose to fit to the Kolmogorov model since the three models described in Sec. 2 converge for frequencies higher than 0.2 Hz and the Kolmogorov model has the simplest form. The method used to gather  $C_n^2$  from the sonic data is described in Sec. 3.2.



**Fig. 9.** a) Representative timing jitter power spectral density used to retrieve  $C_n^2$ . The PSD (blue line) is generated from 1-minute long time series. To evenly weight the fitting in logarithmic frequency space we log-average the PSD (black dots). We then use these points in fitting the Kolmogorov model (red line) between  $0.19 \text{ Hz} < f < 5 \text{ Hz}$ . b) Time series of  $C_n^2$  retrieved from the sonics (gray line) and comb timing jitter data (red circles). Only  $C_n^2$  points for minutes with  $\langle V_{\perp} \rangle > 0.5 \text{ m/s}$  are shown for the comb-retrieved  $C_n^2$ . Below the  $C_n^2$  time series is a time series of the 1-minute averaged  $\langle V_{\perp} \rangle$  from the sonics. c) Ratio of the comb-retrieved  $C_n^2$  to sonic-retrieved  $C_n^2$  as a function of  $\langle V_{\perp} \rangle$ .

A time series of the retrieved  $C_n^2$  from both the sonics (grey line) and frequency comb data (red circles) for June 7th is shown in Fig. 9(b). The comb-retrieved  $C_n^2$  values track the slow trends in  $C_n^2$  such as the gradual rise from 9:00am to 11:00am and the rapid decline from 11:00am to 11:45am. It is also clear that the comb-retrieved  $C_n^2$  is systematically larger than the

sonic-retrieved  $C_n^2$ . Plotting the ratio of comb-retrieved  $C_n^2$  to sonic-retrieved  $C_n^2$  as a function of  $\langle V_\perp \rangle$  helps illuminate this discrepancy (Fig. 9(c)).

The ratio of comb-retrieved  $C_n^2$  to sonic-retrieved  $C_n^2$  is asymptotically high at  $\langle V_\perp \rangle = 0$  m/s, settling into agreement within a factor of  $\sim 5$  for  $\langle V_\perp \rangle > 1$  m/s, and with some indication that agreement may continue to improve with higher wind speed. This points to a breakdown of the model's ability to predict timing jitter at low wind speeds ( $< 1$  m/s). It should be noted that the direct structure function approach to calculate  $C_n^2$  also relies on a wind speed term,  $U$ . However, looking at the one minute averages during our measurement period on June 7<sup>th</sup>,  $U$  was greater than  $\langle V_\perp \rangle$  98% of the time, with  $U$  averaging 1.4 m/s versus 0.7 m/s for  $\langle V_\perp \rangle$  over that period. This systematic difference between  $U$  and  $\langle V_\perp \rangle$  is due to the contribution of  $V_\parallel$  to  $U$ . The higher values of  $U$  would cause the zero-wind-speed asymptotic behavior less frequently in the direct structure function approach than in the optical timing jitter retrieval. There are also interesting questions about the nature of atmospheric turbulence at zero wind speed, for example the breakdown of Taylor's hypothesis that underpins these analyses, but such investigations are beyond the scope of this work.

## 6. Conclusion

In this paper we presented results of pulse timing jitter over a uniform outdoor path due to atmospheric piston noise. Adjacent to the path we installed a meteorological tower with ultrasonic anemometers and a barometer to simultaneously characterize the turbulence. Consistent with previous measurements using a similar frequency-comb laser system [13], we saw a reduced power law at frequencies corresponding to the inertial range. Unlike these previous measurements, however, we see evidence of an outer scale, and in comparing our results to existing theory, we found the Greenwood-Tarazano model best describes the low frequency behavior of the timing jitter power spectral density. In contrast, we found the von Kármán model underestimates the low-frequency power spectrum. Finally, we compared the refractive index structure parameter,  $C_n^2$ , retrieved from the optical pulse timing noise as measured by the frequency comb system to  $C_n^2$  derived from the micrometeorological instruments and found agreement between the two, which improves with increasing wind speed. These results confirm that conventional micrometeorological sensors can accurately estimate timing jitter over nearby paths for wind speeds  $> 1$  m/s.

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## Disclosures

The authors declare no conflicts of interest.

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